TANZANIA HEADS OF ISLAMIC SCHOOL COUNCIL

FORM IV INTER ISLAMIC MOCK EXAMINATION 2024

BIOLOGY 2A

MARKING SCHEME

1. (a) Water turned from colourless to purple colour. Pink, movement of particle **[03 marks]**
   (b) This shows the movement of particles from region of high concentration to a region of low concentration of particles took place. Observe diffusion **[04 marks]**
   (c) Diffusion process **[02 marks]**
   (d) Significance of diffusion to animals:
   (i) It facilitates the absorption of end products of digested food.
   (ii) It facilitates the process of gaseous exchange in animals and human beings.
   (iii) It facilitates the filtration of nutrients and particles in the kidney.
   (iv) It facilitates the uptake of hormones from where they are secreted as they are directly taken by blood. **Any three @ 02 = 06**

   To plants:

   (i) Facilitate minerals salt uptake from soil.
   (ii) Facilitate carbondioxide uptake to the plants tissues. **[Any three @ 02 = 06]**

   Tissues

   (iii) Facilitate movement of hormones and enzymes in plants.

   (e) Organs which perform the process of diffusion are:
   (i) In animals: Kidney and lungs.
   (ii) In plants: Leaves, roots and stem hairs **02 = 04**

2. (a) Identification of specimen U – X.

SPECIMEN	COMMON NAME
U	Spider
V	Beetle
W	Earthworm
X	Maize grain/seed

@ 01 = 04 marks

(b) The differences between U (spider) and V (beetle)

SPECIMEN U	SPECIMEN V
(i) They have two body division (cephalothorax), abdomem	They have three body division head, thorax, abdomen
(ii) They have four pairs of jointed legs	They have three pairs of jointed legs
(iii) They lack wings	They have two or one pair of wings
(iv) They lack antennae	The have one pair of antennae

Any three @ 03 @ 01 ½ = 04 ½

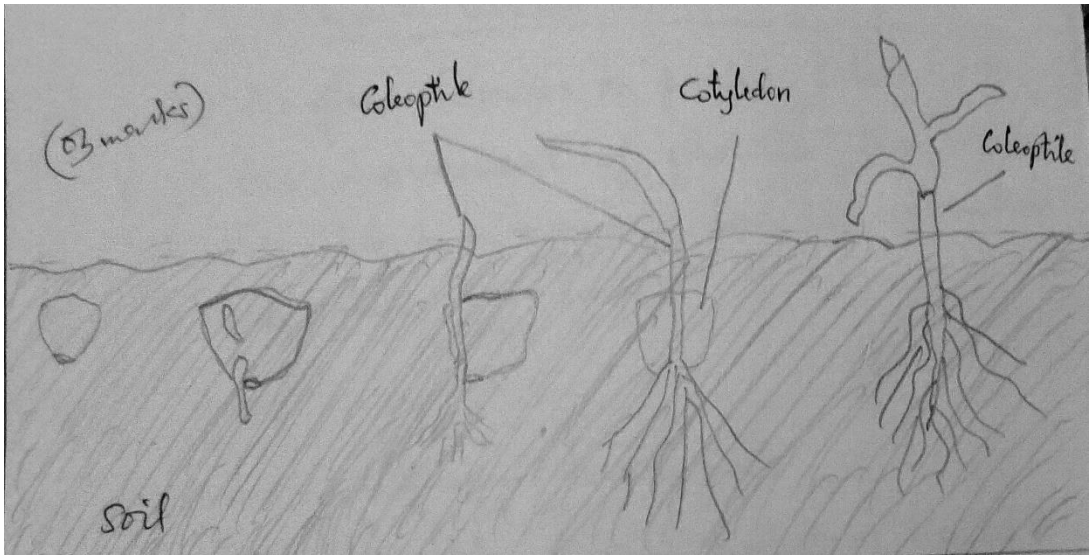
(c) Classification of specimen W (Earthworm)

- ✓ Kingdom : Animalia
- ✓ Phylum : Annelida
- ✓ Class : Oligochaeta **@ 01 = 03**

(d) ECONOMIC importance of Earthworm in Agriculture:

- ✓ They make burrow that facilitate infiltration of water in the ground.
- ✓ They facilitate decomposition of decaying organic matters to release nutrients for plants growth.
- ✓ They contributing the soil air by increasing soil porosity.
- ✓ They facilitate easy penetration of roots through the burrows.
- ✓ Scientific studies
- ✓ Used in fishing bait **Any three @ 1 ½ =4 ½**

(e) The type of gemination of specimen X (maize) is hypogeal germination. **02marks**
Diagram showing hypogeal germination of maize



(f) Adaptation of specimen (Beetle).

- ✓ They posses' variety of locomoting structures such as wing and legs.
- ✓ They have variety of habitat.
- ✓ They have exoskeleton for protection.
- ✓ They feed on variety of food.
- ✓ They have pain of antennae for sensing or nerve coordination.
- ✓ They have compound eyes for sighting.

[Any four @ 01 = 04]

Wabillah Tawfiq